Mathematical Statistics I
HOMEWORK 11

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1.

a.

**could not include but the ~ in the equations is actually $\overset{iid}{\sim}$

$$X_1, \dots, X_5 \sim Uniform(0,15)$$

$$F_X(x) = \frac{x}{15}$$

$$P(M < 10) = P(\max(X_1, \dots, X_5) < 10)$$

$$P(M < 10) = [F_X(10)]^5$$

$$F_X(10) = \frac{10}{15} = \frac{2}{3}$$

$$P(M < 10) = \left(\frac{2}{3}\right)^5 = \frac{32}{243} = 0.132$$

The probability that your longest wait for that week is less than 10 minutes is 0.132.

b.

**could not include but the ~ in the equations is actually $\overset{iid}{\sim}$

$$X_1, \dots, X_5 \sim Uniform(0,15)$$

$$F_X(x) = \frac{x}{15} \quad , \qquad f_X(x) = \frac{1}{15}$$

$$M = \max(X_1, \dots, X_5)$$

$$f_M(m) = 5 f_X(m)[F_X(m)]^4$$

$$f_M(m) = 5\left(\frac{1}{15}\right)\left(\frac{m}{15}\right)^4$$

$$E[M] = \int_0^{15} m\, f_M(m) dm$$

$$E[M] = \int_0^{15} m \cdot 5\left(\frac{1}{15}\right)\left(\frac{m}{15}\right)^4 dm$$

$$E[M] = \frac{5}{15 \cdot 15^4}\int_0^{15} m^5 dm$$

$$\int_0^{15} m^5 dm = \frac{15^6}{6}$$

$$E[M] = \frac{5}{15^5} \cdot \frac{15^6}{6}$$

$$E[M] = \frac{5 \cdot 15}{6} = 12.5$$

$$E[M] = 12.5$$

The average (expected) longest wait for that week is 12.5 minutes.

2.

a.

**could not include but the ~ in the equations is actually $\overset{iid}{\sim}$

$$Y_1, Y_2 \sim f_Y(y) = \frac{1}{100} e^{-y/100} \ , \quad y > 0$$

$$X = \min(Y_1, Y_2)$$

$$P(Y_i > x) = e^{-x/100} \ , \quad x > 0$$

$$P(X > x) = P(Y_1 > x, \ Y_2 > x)$$

$$P(X > x) = P(Y_1 > x)P(Y_2 > x) = e^{-x/100} \cdot e^{-x/100} = e^{-x/50}$$

$$F_X(x) = P(X \le x) = 1 - P(X > x) = 1 - e^{-x/50} \ , \qquad x > 0$$

$$f_X(x) = \frac{d}{dx}F_X(x) = \frac{1}{50}e^{-x/50} \ , \qquad x > 0$$

$$f_X(x) = \frac{1}{50}e^{-x/50} \ , \qquad x > 0$$

The density function for the life of the system is $f_X(x) = \frac{1}{50}e^{-x/50}$ for $x > 0$.

b.

$$f_X(x) = \frac{1}{50} e^{-x/50} \ , \qquad x > 0$$

$$X \sim Exponential\left(\lambda = \frac{1}{50}\right)$$

$$E[X] = \frac{1}{\lambda}$$

$$E[X] = 50$$

The mean length of life for the system is 50 minutes.

3.

a.

$$Y \sim Exponential(mean\ 4)$$

$$f_Y(y) = \frac{1}{4} e^{-y/4} \ , \qquad y > 0$$

$$U = 3Y + 1$$

$$Y = \frac{U - 1}{3}$$

$$\frac{dY}{dU} = \frac{1}{3}$$

$$f_U(u) = f_Y\left(\frac{u-1}{3}\right)\left|\frac{1}{3}\right|$$

$$f_U(u) = \frac{1}{4} e^{-(u-1)/12} \cdot \frac{1}{3}$$

$$f_U(u) = \frac{1}{12} e^{-(u-1)/12} \ , \quad u > 1 \quad and\ 0\ otherwise$$

b.

$$U = 3Y + 1$$

$$E[U] = E[3Y + 1]$$

$$E[U] = 3E[Y] + 1$$

$$E[Y] = 4$$

$$E[U] = 3(4) + 1$$

$$E[U] = 13$$

The expected cost of flour per day is 13 .